MODULE 24 · EXAM STRATEGY - ALL DOMAINS

Confusion Pairs

Almost no new content. Everything here has been taught; the job now is telling them apart at speed.

COVERS All domains NEW CONTENT Almost none MODULE 24 of 27

admin@sthithapragna.com

Wrong answers are rarely wrong. They are usually **the right answer to a slightly different question**.

admin@sthithapragna.com

The one that costs most marks

CLI WORKSPACE

A state file selector

- Separate state inside one working directory
- One configuration, one backend, one credential set
- Switched with terraform workspace select
- Not appropriate for separate credentials

HCP TERRAFORM WORKSPACE

A managed environment

- Its own configuration and working directory
- Its own variables, credentials, permissions
- Its own run history and state versions
- The unit of organisation in HCP Terraform

THE TIEBREAKER

HashiCorp calls these 'distinctly different'. The word appears constantly in questions, so decide which kind is meant before reading the options.

The refresh family

WRITTEN AS	WHAT IT MEANS
`Default plan or apply`	Refreshes everything, then compares against configuration
`-refresh=false`	Skips reading reality. Faster, and the plan may be stale
`plan -refresh-only`	Reads reality and proposes no infrastructure changes
`apply -refresh-only`	Writes those findings into state. Adopts drift
`terraform refresh`	Deprecated since 0.15.4. No preview, no prompt

ON THE EXAM

-refresh=false ignores reality. -refresh-only looks at nothing else. Opposite meanings, one shared word.

Changing state without changing infrastructure

MECHANISM	WHAT IT DOES
`moved block`	Same object, new address. Terraform keeps managing it
`removed, destroy = false`	Terraform stops managing it. The object survives
`removed, destroy = true`	Terraform stops managing it and destroys it
`import block`	An existing object comes under management
`state mv / state rm`	The same jobs, immediately, with no plan and no review

ON THE EXAM

The lifecycle block inside removed is required, and its destroy argument is the difference between a refactor and a deletion.

Ordering, and forcing a rebuild

MECHANISM	WHAT IT ACTUALLY DOES
An attribute reference	Creates the dependency implicitly. Almost always the right answer
depends_on	Adds ordering only. Says after; never says again
replace_triggered_by	Replaces this resource when the referenced thing changes
-replace=ADDRESS	Replaces one instance now, visibly in the plan
terraform taint	Deprecated. Marked state directly with no preview

ON THE EXAM

depends_on says after. replace_triggered_by says again. -replace says now. Three different requests.

The four ways to assert something

MECHANISM	WHEN IT RUNS, AND WHAT FAILURE DOES
Variable validation	When variables are set. Blocks. Since 1.9 can see other variables
Precondition	Before the block is provisioned. Blocks. Cannot use self
Postcondition	After it is provisioned. Blocks. Can use self
Check block	Last step of plan or apply. WARNS, and continues
Sentinel or OPA policy	Between plan and apply. Owned by another team

ON THE EXAM

Only the check block lets the operation continue. Only a policy cannot be removed by the person writing the configuration.

The secrets family

MECHANISM	WHAT IT CONTROLS
`sensitive = true`	Display only. The value is still stored, in plain text
`ephemeral`	Lifetime. Never written to the plan artefact or state
`Write-only argument`	Lets an ephemeral value reach a managed resource. Not persisted
`_wo_version`	IS stored in state. Incrementing it accepts a new value
`Dynamic credentials`	No stored secret at all. Issued per run, expires with it

ON THE EXAM

State, saved plan files and TRACE logs all hold secrets in plain text. sensitive has never affected any of them.

Repeating things

MECHANISM	WHAT IT REPEATS, AND HOW IDENTITY WORKS
`count`	A resource. Identity is a position, from zero
`for_each`	A resource. Identity is a key, from a map or a set of strings
`dynamic block`	A nested block inside one resource. Not the resource itself
`for expression`	Builds a value. Creates no resources at all
`count = 0`	The conditional resource pattern. There is no if statement

ON THE EXAM

count numbers, for_each names. Keys and counts must be known at plan time, because they become addresses in the graph.

Three version numbers, three mechanisms

WHAT IS BEING PINNED	HOW
Terraform itself	required_version, in the terraform block
A provider	version, inside required_providers. Recorded in the lock file
A registry module	version, on the module block. NOT recorded anywhere
A Git-sourced module	The ref argument. version is rejected
A local path module	Nothing. Whatever is on disk is what runs

ON THE EXAM

The lock file records providers only. A module constraint is the entire pinning mechanism, which is why exact pins matter there.

The HCP Terraform family

THE WORD	WHAT IT ACTUALLY IS
Workflow	What starts a run: VCS, CLI or API
Execution mode	Where it runs: Remote, Local or Agent
Local execution mode	The workspace becomes a state store. Policies stop applying
Agent mode	Runs inside your network, and stays an HCP Terraform run
Terraform Enterprise	Self-hosted. The whole platform inside your boundary

ON THE EXAM

Agents move execution. They do not move state. Only Terraform Enterprise keeps state inside your own infrastructure.

Files that sound alike

FILE	COMMITTED?
`.terraform/`	No. A disposable working directory. Recreated by init
`.terraform.lock.hcl`	Yes. Records provider selections and checksums
`terraform.tfstate`	No. And treat it as sensitive data
`terraform.tfvars`	Auto-loaded. Never put secrets in it
`prod.tfvars`	NOT auto-loaded. Needs -var-file
`tfplan`	No. Holds the configuration and secrets in cleartext

> ON THE EXAM
>
> Two names differ by four characters and have opposite treatment: the .terraform directory is ignored, the lock file is committed.

Language pairs worth a second look

PAIR	THE DIFFERENCE
`resource vs data`	Ownership. Ask what should happen on destroy
`variable vs local vs output`	In, stays put, out. That table answers most design questions
`list vs set`	Ordered with duplicates, versus neither. Drives for_each
`try vs can`	try returns a value; can returns a boolean
`file vs templatefile`	file returns contents verbatim; templatefile renders values in
`validate vs plan`	Structure using local schemas, versus reality using credentials

ON THE EXAM

None of these is subtle once stated. All of them are easy to get backwards at speed, which is exactly what the exam measures.

Precedence, in both directions

CONTEXT	WHICH WAY IT RUNS
CLI variables	Weakest to strongest: default, TF_VAR_, tfvars, auto.tfvars, CLI
The trap there	TF_VAR_ is second weakest. Files beat environment variables
HCP variable sets	More specific scope wins. Workspace beats project beats global
The exception	A priority variable set overrides more specific scopes, including -var
Two sets, same scope	Lexical order of the set names. Not the most recently edited

ON THE EXAM

The CLI order and the HCP scope order are different systems. A priority variable set inverts the CLI order entirely.

> A workspace must have its own cloud credentials, its own list of who may apply, and a record of every run for audit.
>
> An engineer says this is what workspaces are for and proposes creating one.

Which kind of workspace satisfies this?

A. A CLI workspace, created with terraform workspace new

B. An HCP Terraform workspace

C. Either. Both isolate credentials and permissions

D. Neither. Credentials and permissions are always organisation-wide

◆ PAUSE AND DECIDE

Which kind of workspace satisfies this?

A A CLI workspace

Isolates state and nothing else. Credentials, permissions and history are all shared with every other workspace in that directory.

B An HCP workspace

Correct. Its own variables, credentials, permissions, run history and state versions is precisely the list of what an HCP workspace holds.

C Either

The assumption the ambiguity exists to catch. HashiCorp calls the two distinctly different.

D Neither

Overcorrects. HCP Terraform's permission model is built on exactly this separation.

THE SIGNAL

Three requirements, three things a CLI workspace does not have. The requirement list is the definition of an HCP workspace.

> An engineer wants to record that somebody resized a database by hand, so Terraform's state matches reality. She must not change any infrastructure.
>
> The configuration still specifies the old size.

Which command?

A terraform apply -refresh=false

B terraform apply -refresh-only

C terraform apply

D terraform refresh

◆ PAUSE AND DECIDE

Which command?

A -refresh=false

The opposite. It skips reading reality, so it cannot record what reality now is.

B -refresh-only

Correct. It compares state against reality, proposes no infrastructure changes, and writes the findings into state once approved.

C Plain apply

Would revert the change, shrinking the database back to the size in the configuration.

D terraform refresh

Deprecated since 0.15.4, and it does this with no preview and no prompt. -refresh-only replaced it.

THE SIGNAL

Adopting drift is only half the job. The configuration still says the old size, so the next normal apply reverts it.

> A storage bucket is being handed to another department. It must keep existing with all its contents, and Terraform must stop managing it.
>
> The change should be reviewed in a pull request like any other.

What should be written?

A. Delete the resource block and apply

B. A moved block pointing at the other department's configuration

C. A removed block with lifecycle destroy set to false

D. prevent_destroy on the resource, then delete the block

◆ PAUSE AND DECIDE

What should be written?

A Delete and apply

Terraform destroys what is no longer declared. The bucket and its contents would go.

B A moved block

moved renames within one configuration's state. It cannot move an object into a different configuration owned by someone else.

C removed, destroy = false

Correct. It forgets the binding, leaves the object alone, and appears in the plan so it goes through review.

D prevent_destroy then delete

The rule dies with the block. Deleting the resource block removes the protection along with it, and the destroy proceeds.

THE SIGNAL

The pair here is moved versus removed. moved keeps a binding and renames it; removed deletes the binding.

> A generated database password must not appear in the state file. The team is on Terraform 1.12 with a current provider.
>
> An engineer has already marked the input variable as sensitive and is surprised the password is still in state.

What actually keeps it out?

A Marking the output sensitive as well

B An ephemeral resource feeding a write-only argument

C Enabling encryption at rest on the backend

D Adding ignore_changes on the password attribute

◆ PAUSE AND DECIDE

What actually keeps it out?

A Sensitive output too

More of the mechanism that already failed. sensitive controls display, not storage, wherever it is applied.

B Ephemeral plus write-only

Correct. Ephemeral values are never persisted, and a write-only argument is the door that lets one reach a managed resource.

C Encryption at rest

Worth doing, and it protects against stolen storage. Anyone with legitimate read access still sees plain text.

D ignore_changes

Stops Terraform proposing updates to an attribute. It does not stop the value being recorded.

THE SIGNAL

sensitive is display, ephemeral is lifetime. This pair appeared in five modules, which is a fair measure of how often it is tested.

> In a CLI-driven run against HCP Terraform, an engineer passes an instance size with -var. The organisation has a priority variable set defining the same key.
>
> She expects her command line value to win, since the command line has the highest precedence.

What happens?

A Her -var value wins. The command line always has highest precedence

B The priority variable set wins. It overrides more specific scopes, including -var

C The run fails with a conflicting variable error

D The values are merged, and the larger size is selected

◆ PAUSE AND DECIDE

What happens?

A **-var wins**

True in the CLI-only precedence order from module eight. A priority variable set is the documented exception that inverts it.

B **The priority set wins**

Correct. Priority sets overwrite variables with the same key at more specific scopes, including command line flags and tfvars files.

C **A conflict error**

Overriding is the intended behaviour throughout Terraform. HCP Terraform shows which value it will use.

D **Merged**

Terraform never merges scalar values. The highest-precedence source simply wins.

THE SIGNAL

Two precedence systems. Module eight's order is correct for the CLI, and a priority variable set inverts it inside HCP Terraform.

> A pipeline runs in an environment with no cloud credentials by design. It must reject a pull request that references an argument name which does not exist on a resource type.
>
> It must not require access to any cloud account.

What should the pipeline run?

A terraform plan, which is the only command that checks resource schemas

B terraform init -backend=false, then terraform validate

C terraform fmt -check -recursive

D A check block asserting the argument exists

◆ PAUSE AND DECIDE

What should the pipeline run?

A plan

Would catch it, and needs credentials and state. The requirement explicitly rules that out.

B init -backend=false, then validate

Correct. validate checks attribute names and types using locally installed schemas, and -backend=false avoids needing the backend.

C fmt -check

A formatter, not a linter. It will happily format a configuration that references arguments which do not exist.

D A check block

Warns rather than blocking, so it could not reject anything. It also runs during a plan, which needs credentials.

THE SIGNAL

validate needs init but not a backend, and no credentials at all. That combination is exactly what a module's CI wants.

Module 24 recap

- **Decide which kind of workspace a question means** before reading the options.
- **-refresh=false ignores reality. -refresh-only looks at nothing else.**
- **moved renames, removed forgets, import adopts.** The destroy argument decides the rest.
- **sensitive is display; ephemeral is lifetime.** Five modules made this point.
- **Only a check block warns. Only a policy cannot be deleted by the author.**
- **Two precedence systems.** The CLI order, and HCP scope with its priority exception.

> ON THE EXAM
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> A distractor is usually a correct statement about something adjacent. Identify the pair, then answer the question you were asked.